Hridayesh Kundu

Goa, India | +91 8902021448 | f20230740@goa.bits-pilani.ac.in | hridaKun.github.io

Electrical and Electronics Engineering student seeking an ASIC Engineer Intern (Implementation) role, with hands-on Verilog RTL design, FPGA synthesis and SystemVerilog verification experience.

EDUCATION

Birla Institute of Technology and Science (BITS) Pilani, K.K. Birla Goa Campus Goa, India
B.E. (Hons.) Electrical and Electronics Engineering and M.Sc. (Hons.) Chemistry | CGPA: 8.96/10 *Expected 2028*

- **Relevant Coursework:** Digital Design, Computer Architecture, Reconfigurable Computing, Microelectronics, Electronic Devices, Microprocessors and Interfacing, Signals and Systems

Vivekananda Academy, Serampore | CBSE Class XII: 87.2% 2022
Gospel Home School | ICSE Class X: 95% 2020

EXPERIENCE

Genesis Gas Solutions Pvt. Ltd. India
Hardware Development Intern *May 2025 – Jul 2025*

- Developed an IoT modem prototype using LoRa communication for wireless data transmission.
- Integrated an Arduino Uno with multiple sensors to implement and validate the communication system on hardware.

PROJECTS

Fixed-Point Quadratic Equation Root Solver – FPGA / RTL Design | *Verilog, Vivado, Zynq-7000* Sep 2026 – Oct 2026

- Designed a hardware solver in Verilog RTL computing real and complex roots of a quadratic equation, with inputs and roots in signed Q16.16 fixed point.
- Implemented iterative Newton-Raphson engines for reciprocal (division) and square root, handling fixed-point truncation and normalization without any vendor arithmetic IP.
- Synthesized and implemented the design in Vivado on a Zynq-7000 FPGA, and performed hardware validation using a VIO-based debug flow to drive coefficients and observe results.

APB Slave Verification Environment – Design Verification | *SystemVerilog* Oct 2026 – Present

- Developing a class-based layered testbench in SystemVerilog for an APB slave, structured as transaction, generator, driver, monitor and scoreboard components connected through mailboxes.
- Targeting constrained-random read/write stimulus checked against a reference memory model, with functional coverage and assertions on the APB handshake as next steps.

Parameterizable UART Transmitter and Receiver – RTL Design | *Verilog, ModelSim* May 2026 – Jul 2026

- Designed parameterizable UART transmitter and receiver modules in Verilog with FSM-based serial logic and configurable parity and stop-bit support.
- Verified functionality using simulation testbenches across multiple transmission scenarios.

Biocoin Glucose Monitoring Analysis – Hardware Research | *Signal Acquisition* Sep 2026 – Present

- Analyzing the sensor front-end and signal acquisition chain of the Biocoin open-source glucose monitoring device to identify hardware limitations and propose accuracy and power improvements.

TECHNICAL SKILLS

Languages (HDL):	Verilog, SystemVerilog, TL-Verilog
EDA Tools:	Xilinx Vivado Design Suite (synthesis, implementation, VIO), ModelSim, Cadence Virtuoso
Hardware:	Zynq-7000 FPGA, Arduino, LoRa
Concepts:	RTL design, FSM design, fixed-point arithmetic, functional verification, computer architecture
Other Tools:	MATLAB, Python, Fusion 360

SCHOLARSHIPS AND HONORS

Samsung Fellowship (Grade II, 75%) May 2026
Awarded for the ISWDP programme conducted by Synopsys, IISc Bangalore and Samsung
Merit-cum-Need Scholarship (80%) Jul 2026
Awarded on the basis of academic merit and annual family income

ACTIVITIES AND LANGUAGES

Chess: Represented BITS Goa in inter-campus chess tournaments, including Arena (BITS Hyderabad) and Spree (BITS Goa).
Languages: English, Hindi, Bengali